Step 0: why the leader always falls

rise-and-fall in the pure-growth model, derived

M. Heltberg

Stepwise-progression note, June 2026

Folder `stepwise_progression/`. Code: `step0_growth.py` (base model), `step0_turnover.py` (the measurements and figure below).

Abstract

The base model has no wars, alliances, or decisions — only resource-limited growth of N states. Empirically a single state dominates for a while and is then *always* overtaken: the condensate is dynamic. We show this analytically. Once the shared resource is exhausted the strength of any state, in particular the leader, obeys a driftless equation $\mathrm{d}S = \sigma\sqrt{S}\,\mathrm{d}W$, which after $S \mapsto 2\sqrt{S}$ is a Bessel process of dimension 0 (equivalently S is a squared-Bessel $\textsc{besq}(0)$). Such a process is a nonnegative martingale that reaches 0 almost surely; optional stopping gives the sharp statement that, from its current size S_0, the leader reaches a level $a > S_0$ before collapsing only with probability S_0/a. Permanent dominance is therefore impossible. We also identify the structural cause — *additive* (size-independent) growth, whose drift-to-noise ratio falls as $S^{-1/2}$ — and contrast it with multiplicative growth, which locks in.

1 Model

N states with strengths $S_i \geq 0$ and growth rates γ_i. The only coupling is the global resource availability R:

$$R(t) = \max\Big(1 - \tfrac{1}{K}\textstyle\sum_j S_j,\, 0\Big), \tag{1}$$

$$\mathrm{d}S_i = \gamma_i\, R\, \mathrm{d}t + \sigma\sqrt{S_i}\,\mathrm{d}W_i, \tag{2}$$

$$\mathrm{d}\gamma_i = \lambda_g(\mu_g - \gamma_i)\,\mathrm{d}t + \sigma_g\,\mathrm{d}W_i^g. \tag{3}$$

Two features matter below: the growth drift $\gamma_i R$ is *additive* (independent of S_i), and the noise is *demographic*, of amplitude $\sigma\sqrt{S_i}$ (Itô; $\sigma =$ `sig_s`). Defaults: $N = 10$, $K = 10N$, $\sigma = 1$.

2 The phenomenon

Measured over a run of length $t = 200$ (Fig. 1a): the instantaneous leader $\arg\max_i S_i$ changes ~ 55–65 times, 6–10 distinct states hold the lead, and the leader at mid-run is still leading at the end only $\approx 42\%$ of the time. No state stays on top.

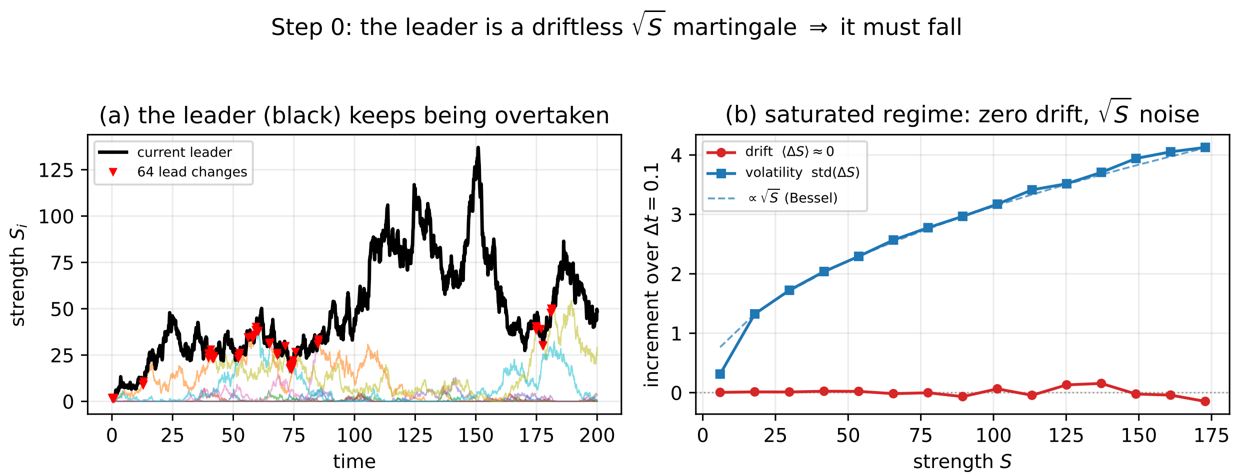

Figure 1: (a) One run: the instantaneous leader (black) is repeatedly overtaken (red markers). (b) Pooled over 40 runs in the saturated window $t > 70$: the conditional drift $\langle \Delta S \rangle$ is ≈ 0 at every size (red), while the conditional volatility grows as $\sqrt{S}$ (blue, with the $\propto \sqrt{S}$ guide). This is the signature of the driftless $\sqrt{S}$ martingale derived in §4.

3 Saturation: the drift vanishes

Summing (2), the total $\Sigma = \sum_i S_i$ has drift $R \sum_i \gamma_i$. While $R > 0$ and $\sum_i \gamma_i > 0$ the total grows, so Σ rises toward K and $R \downarrow 0$; the resource is exhausted (in the defaults by $t \approx 70$). Once there, $R \approx 0$ and (2) reduces, for *every* state, to

$$\mathrm{d}S \;\simeq\; \sigma\sqrt{S}\,\mathrm{d}W. \tag{4}$$

The drift is then 0 *regardless of size*: this is exactly what Fig. 1b shows ($\langle \Delta S \rangle \approx 0$ across all S). Equation (4) governs the leader.

4 The leader is a Bessel martingale

Martingale. Equation (4) has zero drift, so S_t is a (local) martingale: $\mathbb{E}[S_{t+s} \mid \mathcal{F}_t] = S_t$ as long as it has not been absorbed. A nonnegative martingale converges almost surely; we now show the limit is 0.

Reduction to a Bessel process. Let $Y = 2\sqrt{S}$, so $S = Y^2/4$. With $Y'(S) = S^{-1/2}$ and $Y''(S) = -\frac{1}{2}S^{-3/2}$, Itô's formula applied to (4), using $(\mathrm{d}S)^2 = \sigma^2 S\,\mathrm{d}t$, gives

$$\mathrm{d}Y = Y'\,\mathrm{d}S + \tfrac{1}{2}Y''(\mathrm{d}S)^2 = S^{-1/2}\,\sigma\sqrt{S}\,\mathrm{d}W - \tfrac{1}{4}\sigma^2 S^{-1/2}\,\mathrm{d}t = \sigma\,\mathrm{d}W - \frac{\sigma^2}{2Y}\,\mathrm{d}t, \tag{5}$$

where we used $S^{-1/2} = 2/Y$. Comparing with the Bessel SDE $\mathrm{d}Y = \frac{\delta-1}{2Y}\mathrm{d}t + \sigma\,\mathrm{d}W$ identifies the dimension

$$\frac{\delta-1}{2} = -\frac{1}{2} \quad\Longrightarrow\quad \boxed{\delta = 0}. \tag{6}$$

Equivalently, a squared-Bessel besq(0). The cleanest statement avoids the transform. Time-change (4) by $s = \frac{\sigma^2}{4}t$ and set $B_s = \frac{\sigma}{2}W_t$ (a standard Brownian motion in s, since $\operatorname{Var} B_s = \frac{\sigma^2}{4}t = s$). Then $\sigma\sqrt{S}\,\mathrm{d}W = 2\sqrt{S}\,\mathrm{d}B$ and (4) becomes

$$\mathrm{d}S = 2\sqrt{S}\,\mathrm{d}B = \underbrace{\delta}_{=0}\,\mathrm{d}s + 2\sqrt{S}\,\mathrm{d}B, \tag{7}$$

which is the squared-Bessel process $\mathrm{BESQ}(\delta)$ of dimension $\delta = 0$. Standard facts: for $\delta < 2$ the origin is reached almost surely, and for $\delta \leq 0$ it is *absorbing*. Hence

Proposition 1. *The driftless strength (4) reaches 0 in finite time with probability one. The leader is absorbed.*

How fast the lead is lost (optional stopping). The martingale property gives a sharp, elementary bound without invoking Bessel theory. Fix the leader's current size S_0 and a target $a > S_0$, and stop at $\tau = T_0 \wedge T_a$ (first hitting of 0 or of a). Since S is a bounded martingale up to τ, optional stopping yields $\mathbb{E}[S_\tau] = S_0$, i.e.

$$a\,\mathbb{P}(T_a < T_0) + 0\cdot\mathbb{P}(T_0 < T_a) = S_0 \implies \boxed{\mathbb{P}(\text{reach } a \text{ before } 0) = \frac{S_0}{a}}. \tag{8}$$

So the chance the leader ever grows to $a = 2S_0, 5S_0, 10S_0$ before collapsing is $\frac{1}{2}, \frac{1}{5}, \frac{1}{10}$; letting $a \to \infty$, $\mathbb{P}(\text{escape to } \infty) = 0$, consistent with Proposition 1. Large further rises are rare and collapse is generic: *rise-and-fall is the typical history of every state.* When the leader dips, $\Sigma < K$ lifts $R > 0$ and the freed resource is captured by another state, which becomes the next (temporary) leader.

5 Why additive growth is the cause

The instability is specific to the *additive* drift γR. Compare the two natural growth laws by their drift-to-noise ratio at size S (drift μ, noise standard rate $\nu = \sigma\sqrt{S}$):

$$\text{additive } (\mu = \gamma R): \quad \frac{\mu}{\nu} = \frac{\gamma R}{\sigma\sqrt{S}} \propto S^{-1/2}, \qquad \text{multiplicative } (\mu = \gamma R\,S): \quad \frac{\mu}{\nu} = \frac{\gamma R\sqrt{S}}{\sigma} \propto S^{+1/2}. \tag{9}$$

For additive growth the ratio *falls* as the state grows: the bigger the leader, the more noise-dominated it is. For multiplicative growth it *rises*: large states are drift-dominated and stable. The same statement in log-strength $y = \ln S$: by Itô, multiplicative growth gives $\mathrm{d}y = (\gamma R - \frac{\sigma^2}{2S})\mathrm{d}t + \frac{\sigma}{\sqrt{S}}\mathrm{d}W$, whose noise $\propto S^{-1/2} \to 0$ — log-strength becomes deterministic and the leader locks in. Empirically, replacing $\gamma R \to \gamma R\,S$ cuts the number of leadership changes by $\sim 5\times$ and makes the mid-run leader stay on top. *Additive, size-independent growth $\Rightarrow$ churn; multiplicative $\Rightarrow$ lock-in.*

6 A numerical caveat

The positivity clamp `S = max(S,0)` reflects the Euler step at 0 and thereby injects a little strength each time (2) would send S negative. Consequently Σ is not conserved and slowly inflates above K for $t \gg 200$, partly masking the $\mathrm{BESQ}(0)$ absorption. The analysis above concerns the clean saturated window $70 \lesssim t \lesssim 200$. For long-time quantitative work use a positivity-preserving CIR scheme (full-truncation / Alfonsi) or enforce the Feller condition $2\gamma R \geq \sigma^2$.

7 Outlook: Step 1 (decay) should set a finite lifetime

Turning on the decay term $-\delta S$ (defined but unused in the base code) makes the strength a genuine *CIR* process,

$$\mathrm{d}S = \big(\gamma R - \delta S\big)\mathrm{d}t + \sigma\sqrt{S}\,\mathrm{d}W, \tag{10}$$

i.e. mean-reverting with rate $\kappa = \delta$ toward $\theta = \gamma R/\delta$. This is qualitatively different from $\textsc{besq}(0)$: the linear restoring drift gives a *stationary* law — the Gamma distribution $S_\infty \sim \Gamma\big(\frac{2\kappa\theta}{\sigma^2}, \frac{2\kappa}{\sigma^2}\big) = \Gamma\big(\frac{2\gamma R}{\sigma^2}, \frac{2\delta}{\sigma^2}\big)$ — and a finite relaxation time $1/\kappa = 1/\delta$. The Feller boundary classification (0 inaccessible iff $2\gamma R \geq \sigma^2$) then controls whether states can die at all. **Prediction to test in Step 1:** decay replaces the unbounded martingale wandering by bounded fluctuations about $\theta = \gamma R/\delta$, giving the leader a characteristic *lifetime* $\sim 1/\delta$ and a tunable rise-and-fall period, rather than the scale-free martingale collapse of Step 0. (Note R is itself dynamic: decay frees resource, so the system settles at a total below K with $R > 0$ sustained, which keeps $\theta > 0$.)